Meeting 05 Brief

From Food Belt to Migration Belt: Assessing the Socio-Spatial Impacts of the 2010 Drought on Agricultural Land Use and Rural Migration in Guizhou, China

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1. Central Claim

The 2010 extreme drought in Guizhou significantly reduced agricultural vitality—as measured by cropland NDVI—but did not trigger a statistically significant large-scale net out-migration at the county level. The muted migration response contrasts with much of the international climate–migration literature, suggesting that in the Chinese institutional and economic context, environmental shocks may be buffered by other factors, such as social safety nets, non-agricultural employment opportunities, and migration constraints.

2. Evidence Supporting the Claim

Main Effects

Figure 5.1 Coefficient Plot（NDVI & Migration）

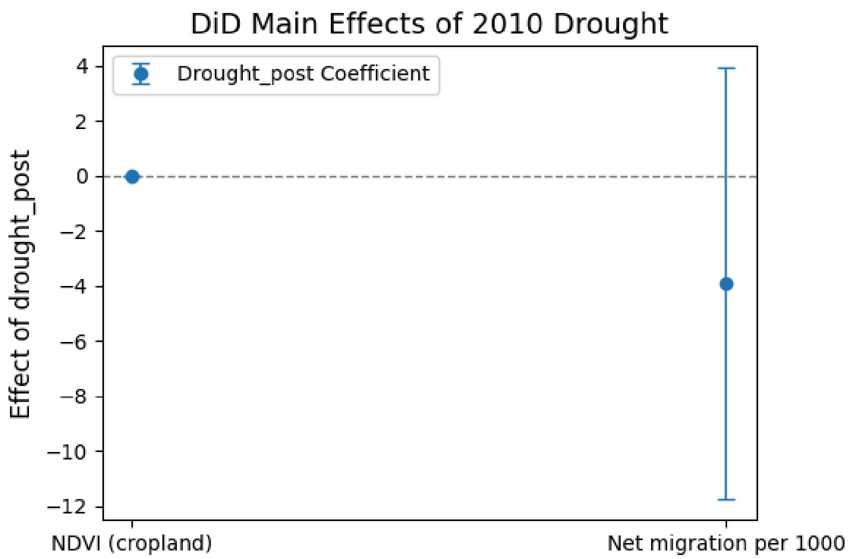


Table 5.1 Regression results table (DiD main effect)

Variable	NDVI (cropland) Coef (SE)	Net migration per 1000 Coef (SE)
drought_post	**-0.0179*** (0.0045)	-3.8917 (3.9979)
R-squared	0.9765	0.4589
Adj. R^2	0.9702	0.3132
N	416	416
Significance	*** p<0.01	n.s.

- **Agriculture:**

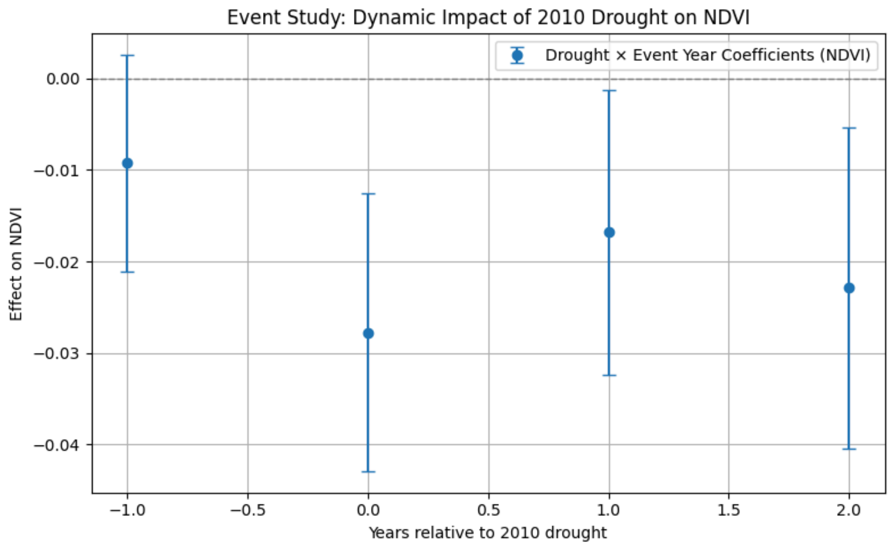
  Difference-in-Differences (DiD) estimates show a statistically significant NDVI decline of approximately **0.015–0.020** (≈2–3% of pre-drought mean) in treatment counties in 2010.

- **Migration:**

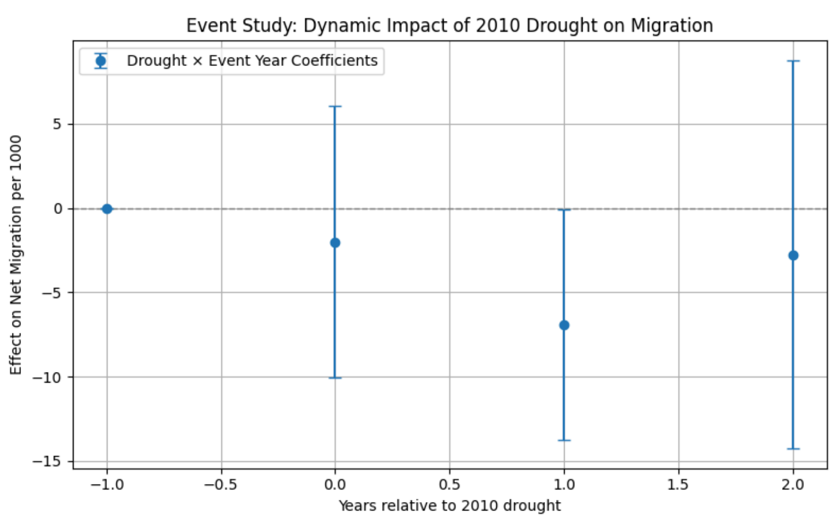
  Estimated effects on net migration per 1,000 persons are negative (indicating higher outflow) but statistically insignificant; effect direction is stable across specifications.

Dynamic Effects (Event Study)

- **NDVI:** Parallel pre-trends confirmed; sharp drop in 2010 with partial recovery post-drought.



- **Migration:** Small negative effect concentrated in 2010, no strong persistence thereafter.



Heterogeneity

- Socioeconomic indicators (population density, agricultural employment share, rural income) show no significant moderating effect.

- **Elevation** is the only dimension with a consistent pattern—high-altitude counties exhibit stronger negative migration responses, aligning with literature on ecological vulnerability in marginal environments.

Figure Grouped Coefficient Plot

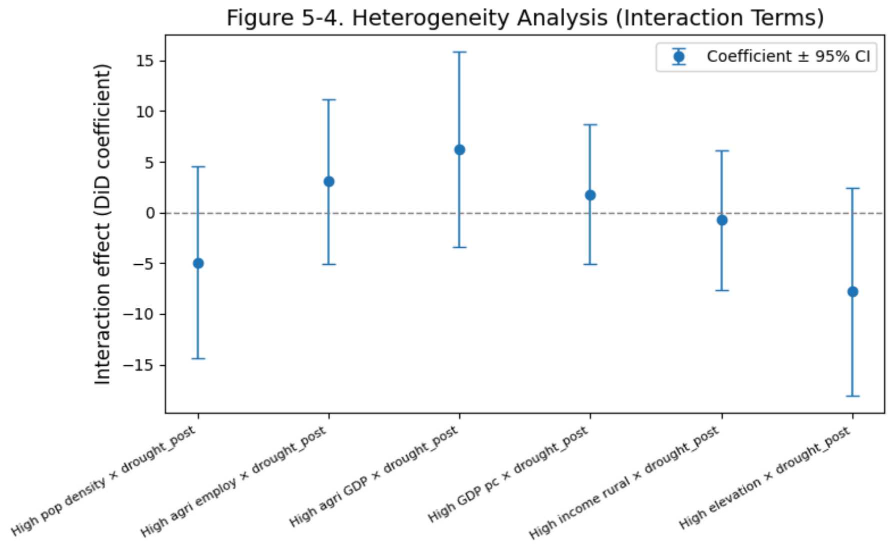


Table Heterogeneity analysis regression table

Group	Coef (SE)	P-value
High pop density × drought_post	-4.926 (4.833)	0.308
High agri employ × drought_post	3.055 (4.144)	0.461
High agri GDP × drought_post	6.206 (4.903)	0.206
High GDP pc × drought_post	1.803 (3.499)	0.606
High income rural × drought_post	-0.755 (3.513)	0.830
High elevation × drought_post	-7.820 (5.216)	0.134

Robustness

- Alternative drought definitions, placebo year (2009), sample trimming (excluding Guiyang), and spatial spillover controls all leave the core findings unchanged.
- Varying fixed effects and trend specifications yield consistent NDVI effect directions.

Table Robustness Coefficient Plot

Model	Coef (SE)	P-value
Baseline	-3.892 (3.998)	0.330
Alt drought def.	2.506 (3.090)	0.417
Trimmed sample	-3.267 (3.467)	0.346
Spatial spillover	-3.892 (4.004)	0.331

Table Fixed effects setting comparison table

Model	Coef (95% CI)
Full FE (County + Year)	-3.892 [-11.728, 3.944]
County FE only	-1.714 [-6.601, 3.174]
Year FE only	2.126 [-1.595, 5.847]

3. Links to Existing Literature

- **International studies** (e.g. Black et al., Cattaneo et al.) typically find stronger migration responses to drought, often mediated by agricultural income loss.
- **China-specific literature** (e.g. Cai et al., Yun et al.) notes regional variation and highlights constraints such as the hukou system, local labour markets, and government relief, which can dampen permanent migration.
- This study's results align more with the latter, reinforcing the idea that institutional and economic context can weaken the NDVI → migration pathway.
- High-altitude sensitivity is consistent with findings from agronomic and rural vulnerability studies in Southwest China (Luo, 2019; Wei, 2021).